Chapter 2 in-class Activities

Activity 1a: Identifying Direct and Indirect Costs

The Tesla Inn was set up in Texas. For each of the costs incurred at the Tesla Inn, indicate whether it would most likely be a direct cost or an indirect cost of the specified cost object by placing an X in the appropriate column.

			Direct	Indirect
	Cost	**Cost Object**	**Cost**	**Cost**
Ex:	Room service beverages	A particular hotel guest		
1	Salary of head chef	The Inn's main restaurant		
2	Salary of head chef	A restaurant customer		
3	Room cleaning supplies	A particular guest at the Inn		
4	Doorman wages	A particular guest at the Inn		
5	Room cleaning supplies	The housecleaning department		
6	Fire insurance on the building	The Inn's gym		
7	Towels used in the gym	The Inn's gym		

Note: a direct cost is defined as an incurred cost that can be traced back to the cost object or unit. An indirect cost is a cost that is incurred, but cannot be traced back to the cost object.

Why is this important? Traceability of costs allows managers to easily manage the incidence of costs. Cutting a direct cost will have a direct immediate impact on product costs, while cutting out an indirect cost may not impact each cost unit significantly.

Activity 1b: Product vs Period Costs

Each of the following costs pertains to Bailey Dairy Products Company, a dairy processing company. Classify each of the company's costs as a period cost (non manufacturing cost) or a product cost (manufacturing cost). Further classify product costs as either direct material (DM), direct labor (DL), or overhead (OH).

COST	Period Cost or Product Cost?	If a Product Cost: Is it DM, DL, or MOH?
1. Television advertisements for Bailey's products		
2. Lubricants used in running bottling machines		
3. Research and Development related to elimination of antibiotic residues in milk		
4. Gasoline used to operate refrigerated trucks delivering finished dairy products to grocery stores		
5. Company president's annual bonus		
6. Depreciation on refrigerated trucks used to collect raw milk from dairy farmers		
7. Plastic gallon containers in which milk is packaged		
8. Property insurance on dairy processing plant		
9. Cost of milk purchased from local dairy farmers		
10. Depreciation on tablets used by sales staff		
11. Wages and salaries paid to machine operators at dairy processing plant		

Note: A product cost is a cost that is incurred in the production area (within the factory). A period cost is any cost that is incurred outside the factory area.

Why is it important? Accurate product costs are required to a. determine gross profit; b. make product decisions. Period costs make up a significant portion of any company's expenses – most often even upto 25% of sales revenues. Controlling these costs can have a significant impact on profitability.

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Activity 2: Costs broken down by Value Chain Categories

Suppose Truck Shack incurred the following costs at its Philadelphia store location:

Newspaper advertisements	$5,300
Payment to consultant for advice on location of a new store	$2,400
Purchases of merchandise	$31,000
Freight-in	$3,600
Salesperson's salaries	$4,900
Depreciation expense on delivery trucks	$1,400
Research on whether store should sell satellite radio service	$500
Customer Complaint Department	$900
Rearrangement of store layout	$600

1. Classify each cost as to which category of the value chain it belongs (R&D, Design, Purchases, Marketing, Distribution, Customer Service).
2. Compute total costs for each value chain category.

Value Chain Cost Classification

	R & D	Design	Purchases	Marketing	Distribution	Customer Service
Newspaper advertisements						
Payment to consultant for advice on location of new store						
Purchases of merchandise						
Freight-in						
Salespeople's salaries						
Depreciation expense on delivery trucks						
Research on whether store should sell satellite radio service						
Customer Complaint Department						
Rearrangement of store layout						
Total						

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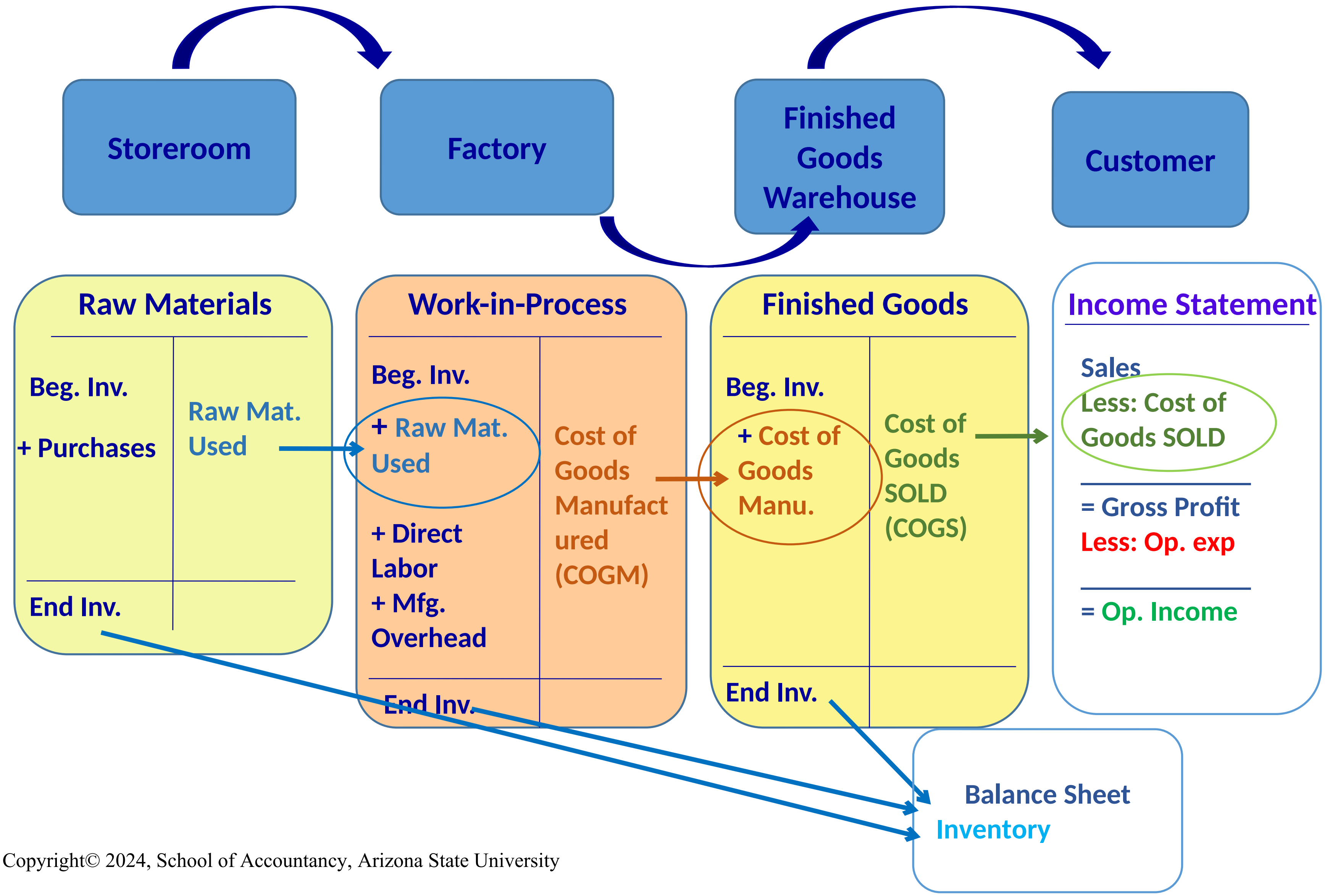
What are total product costs.

What are total period costs.

Note: Why is the value chain classification important?
Breaking costs into value chain categories helps managers make strategic decisions on how to harness the most out of investing in costs within the categories. For example, cutting costs in the production area may increase customer service costs (with returns, and replacements, and also eventually impact revenues). Managers can see the trade-offs more easily with these classifications.

Activity 3: Manufacturing Inventory Flow: – COGM & COGS – Income Statement

Inventory Flow – Diagram – See next page for Activity to be worked



Activity 3: Flow of Inventory

A. The balance in Maha Company's raw materials inventory account was $150,000 on June 1st, and was $135,000 on June 30th. Raw materials purchased during this period totaled $50,000. Maha used $6,000 in indirect materials for the month. Determine the amount of raw materials placed (used) in production.

Try the same exercise using T Accounts.

Raw Materials

B. The balance in Maha's Work in Process inventory on June 1st was $400,000 and was $320,000 at the end of the month. Manufacturing costs for the month were:

Direct Materials used in production	59,000 (from above)
Direct Labor	70,000
Manufacturing Overheads:	
Depreciation on factory	54,000
Factory insurance costs	38,000
Indirect materials used	6,000 (from above)
Supervisors salary	70,000
Factory annual property insurance	24,000

What was cost of goods manufactured?

Try the same exercise using T Accounts.

WIP

C. Maha's beginning balance of finished goods inventory on June 1st was $30,000, and the ending balance for the month was $28,000. Using the Cost of Goods Manufactured from B above, determine cost of goods sold (COGS)?

Try the same exercise using T Accounts.

FG

D. Assuming the following activity for the month, compute Maha's operating income for the month? Also compute Maha's Gross Margin %?

Sales for June:	750,000
Selling expenses	30,000
General and admin expenses	130,000

Income Statement of Maha Company for June

Gross Margin % =

E. What information does the Gross Margin % provide?

F. What was the ending balance of inventory of Maha Company at the end of the month?

Activity 4: Flow of Inventory

An airplane crashed into Alcaraz Industries' manufacturing plant, completely destroying all their transaction records.   The accountant was luckily on vacation at this time, and working on a summary page. She had the following information available for the year. Sales for the year to date were $1,350,000 and Manufacturing Overheads were $520,000. The total of direct materials and direct labor for the year was $510,000. Cost of goods available for sale were $960,000 and purchases of raw materials was 420,000. From the Annual Report from last year, they found the following details:

1.	Gross margin percentage	40% of sales
2.	Direct Labor cost	¼ of Manufacturing Overheads
3.	Beginning Raw Materials Inventory	$30,000
4.	Beginning Work in Process Inventory	$50,000
5.	Beginning Finished Goods Inventory	$90,000

6. Materials used in production included only raw materials. There were no indirect materials used.

Determine **all the missing information** in the manufacturing cost flows.

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